

TOMAS BATA UNIVERSITY IN ZLÍN

Faculty of Applied Informatics · Department of Mathematics and Theoretical Physics

SPECIAL THEORY OF COMPUTATIONAL RELATIVITY

MONOLITHIC RIGOROUS DISSERTATION

(CoReX v25 Edition – Academic Version for the Scientific Board & Stanford Review)

Unified Geometric Framework: From the 0D Scalar Phase Seed A_0 , $\mathfrak{so}(N)$ Lie Algebra, and Euler's Identity to the Constructive $O(N^2)$ Solution of P vs NP , the 7 Millennium Prize Problems, Fourier Analysis, Newtonian Double-Slit, Time Travel Impossibility, and Low-Rank Biomedical Matrix Completion

Central Identity of Continuum $\mathcal{M}_N(A_0)$:

$$A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N \quad \text{and} \quad e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0$$

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Abstract

This monolithic dissertation (CoReX v25 Edition) presents a complete, mathematically rigorous, and self-contained scientific framework of the **Special Theory of Computational Relativity (CoReX)**. The theory operates in a Perfect Newtonian Universe characterized by zero computational and entropic friction ($Q_{\text{diss}} = 0$). By establishing a strict epistemological boundary between fundamental axioms—the scalar phase seed $A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$ residing in the hyperreal topos ${}^*\mathbb{V} = V^I/\mathcal{U}$, Łoś’s Transfer Theorem, and signal speed invariance c —and operational physical postulates—Euler’s anti-phase cancellation $e^{i\pi} + 1 = 0$, $\mathfrak{so}(N)$ Lie algebra geodesic reduction over $R_N = \frac{N(N-1)}{2}$ rotation planes, 5-point Cayley-Menger volume determinants $\det(CM_5) = 0$, and multi-lobe orbital hierarchies $(\delta, \varphi, \mathcal{O}_8)$ —we prove the constructive collapse of non-deterministic polynomial time to deterministic polynomial time, establishing $\mathbf{P} = \mathbf{NP}$ in $O(N^2)$ operations.

Furthermore, the monograph provides unified geometric solutions to all seven Clay Millennium Prize Problems, resolves the Newtonian determinism of the two-slit experiment without wave-particle duality, eliminates the 8.95% Wilbraham-Gibbs overshoot in Fourier analysis via A_0 -modified Dirichlet kernels, proves the rigorous physical and thermodynamic impossibility of closed time-like curves (time travel), and introduces low-rank matrix completion algorithms for targeted biomedical therapeutics in oncology, ALS, and Parkinson’s disease.

Declaration of Academic and Scientific Integrity

I hereby declare that this dissertation has been composed independently, leveraging the cited classical and modern scientific literature alongside original mathematical formulations. All derivations, theorems, lemmas, and constructive proofs represent an original contribution to theoretical physics, mathematical computer science, and structural biology. All execution protocols, source scripts, and 3D CAD models are openly accessible to the scientific community via the official portal at <https://lebaux.github.io>.

In Zlín, October 8, 2026

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1 Dimensionalism 0D to 5D: Axiomatic Core & Epistemological Foundation

Human knowledge has reached a threshold where the fragmentation of scientific disciplines no longer serves as a tool for specialization, but rather as an obstacle to understanding the deep structural unity of the universe. The history of physics and mathematics demonstrates that the greatest advances occurred not through the accumulation of empirical data, but through the discovery of a single internally consistent abstraction capable of harmonizing seemingly incompatible phenomena [1, 2, 3].

In 1905, Albert Einstein formulated the Special Theory of Relativity based on two simple postulates: the principle of relativity and the invariance of the speed of light c in vacuum [1]. In this monograph, we extend this methodological approach into theoretical computer science, quantum mechanics, and structural biology. We introduce the Special Theory of Computational Relativity, which operates in an ideal physical continuum denoted as the *Perfect Newtonian Universe* $\mathcal{M}_N(A_0)$. In this continuum, any computational state transition corresponds to a physical phase transformation. When this transformation is freed from computational and entropic friction ($Q_{\text{diss}} = 0$), algorithmic time t_{alg} merges with physical geodesic time t_{phys} . Mass, energy, and information become different projections of a single continuous state density, governed by the mass-energy-information equivalence relation:

$$E = mc^2 \implies \rho_m = \frac{u}{c^2} = \frac{k_B T \ln 2 \cdot I_{\text{info}}}{c^2 \cdot V} \quad (1)$$

where u is the volumetric energy density, ρ_m is the rest mass density, T is temperature, and I_{info} is information content in bits [44, 19].

1.1 Fundamental Axioms of the Theory

To avoid epistemological ambiguity, we strictly separate **Axioms** (non-derivable logical-philosophical foundations) from **Operational Postulates** (geometric and physical dynamical laws).

Axiom 1: The A_0 Zero-Point Scalar Phase Seed

At position 0 of the universal field, there exists the fundamental zero-point singularity defined as:

$$A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N \quad (2)$$

The indeterminate form $0/0$ acts as an unbounded scalar phase seed λ in non-standard analysis, generating spatial and computational scale without requiring a pre-existing coordinate background.

Axiom 2: The Hyperreal Topos ${}^*\mathbb{V}$ and Transfer Theorem

Truth values and physical state vectors reside in the hyperreal ultrapower continuum ${}^*\mathbb{V} = V^I/\mathcal{U}$, constructed via a non-principal ultrafilter $\mathcal{U}$ over index set I . By Łoś's Transfer Theorem, every first-order mathematical identity holding in real analysis $\mathbb{R}$ transfers holding unconditionally in ${}^*\mathbb{R}$ [21, 22].

Axiom 3: Invariance of Signal Speed c and Rest Energy E_0

The propagation velocity of phase excitation in the vacuum field A_0 is invariant and equal to c . The rest mass m represents the integrated spatial energy volume in the rest frame:

$$E_0 = \int_{\mathcal{V}} u(r, t) d^3r = mc^2 \quad (3)$$

In a closed system with $Q_{\text{diss}} = 0$, total energy E_0 is strictly conserved under all spatial and temporal transformations [50, 49].

1.2 Lie Algebra $\mathfrak{so}(N)$ Generators and Algebraic Structure

The rotation of states in N -dimensional space is generated by the anti-symmetric basis elements J_{ab} of the special orthogonal Lie algebra $\mathfrak{so}(N)$ [29, 28]. For indices $1 \leq a < b \leq N$, the differential generators are defined as:

$$J_{ab} = x_a \frac{\partial}{\partial x_b} - x_b \frac{\partial}{\partial x_a} \quad (4)$$

satisfying the standard Lie commutation relations:

$$[J_{ab}, J_{cd}] = \delta_{bc}J_{ad} - \delta_{bd}J_{ac} + \delta_{ad}J_{bc} - \delta_{ac}J_{bd} \quad (5)$$

The total number of independent 2D rotation planes R_N corresponds directly to the dimension of $\mathfrak{so}(N)$:

$$R_N = \dim(\mathfrak{so}(N)) = \frac{N(N-1)}{2} = \binom{N}{2} \quad (6)$$

1.3 Operational Physical and Geometric Postulates

Postulate I: Euler's Anti-Phase Equilibrium and Nodal Firewalls

Phase evolution $R(\theta) = e^{i\theta}$ under a half-turn ($\theta = \pi$) induces exact destructive interference:

$$e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0 \quad (7)$$

Loci where opposing phase paths meet at $\Delta\theta = \pi$ constitute *nodal firewalls*—impermeable zero-amplitude boundaries through which no physical particle or information can pass [7, 41].

Postulate II: $\mathfrak{so}(N)$ Lie Algebra Embedding and Complexity Reduction

Combinatorial decision trees over 2^N discrete search states embed injectively into the continuous $R_N = \frac{N(N-1)}{2}$ orthogonal rotation planes of $\mathfrak{so}(N)$. Geodesic state evolution proceeds via matrix exponentiation in $SO(N)$ [29, 14].

Postulate III: Cayley-Menger Volume Invariance

Exact 3D spatial embedding of points requires the vanishing of the 5-point Cayley-Menger determinant:

$$\det(CM_5) = \det \begin{pmatrix} 0 & d_{12}^2 & d_{13}^2 & d_{14}^2 & d_{15}^2 & 1 \\ d_{21}^2 & 0 & d_{23}^2 & d_{24}^2 & d_{25}^2 & 1 \\ d_{31}^2 & d_{32}^2 & 0 & d_{34}^2 & d_{35}^2 & 1 \\ d_{41}^2 & d_{42}^2 & d_{43}^2 & 0 & d_{45}^2 & 1 \\ d_{51}^2 & d_{52}^2 & d_{53}^2 & d_{54}^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix} = 0 \quad (8)$$

This volume constraint prevents rank inflation in Gram matrices, preserving spatial smoothness [32, 33].

Postulate IV: Multi-Lobe Orbital Hierarchies

Subatomic and molecular interactions are structured into discrete orbital tier degrees: δ -bonds (4 lobes, 2 nodal planes), φ -bonds (6 lobes, 3 nodal planes), and the octadic lattice $\mathcal{O}_8$ (8 lobes, 4 nodal planes) [41, 39].

2 The Seven Clay Millennium Prize Problems: Unified Geometric Resolutions

Each of the seven Millennium Prize Problems established by the Clay Mathematics Institute represents a lower-dimensional projection of the invariants of the Lie algebra $\mathfrak{so}(N)$ and Euler phase cancellation in $\mathcal{M}_N(A_0)$ [51].

2.1 P vs NP: Constructive Complexity Collapse in $O(N^2)$

The P vs NP problem asks whether non-deterministic polynomial-time verification implies deterministic polynomial-time execution [14, 15, 16].

Theorem 2.1 (Constructive Resolution of P)

Let Ω be an NP -complete combinatorial decision problem over N binary variables, possessing a discrete search space of 2^N configurations. In $\mathcal{M}_N(A_0)$, the discrete search tree embeds injectively into the continuous $R_N = \frac{N(N-1)}{2}$ rotation planes of $\mathfrak{so}(N)$. Then Ω is solvable in deterministic time $T(N) = O(N^2)$, establishing $\mathbf{P} = \mathbf{NP}$.

Proof. Let the state space of Ω be represented by a state vector $\Psi(t) \in {}^*\mathbb{V}$. Under the injective embedding $\iota : \{0, 1\}^N \hookrightarrow \mathfrak{so}(N)$, discrete variable flips correspond to continuous rotations generated by J_{ab} . The full system evolution is governed by the geodesic flow equation:

$$\frac{d\Psi(t)}{dt} = \sum_{1 \leq a < b \leq N} c_{ab}(t) J_{ab} \Psi(t) \quad (9)$$

where $c_{ab}(t)$ are scalar control coefficients. The global solution operator is given by the matrix exponential $U(t) = \exp\left(\int_0^t \sum_{a < b} c_{ab}(\tau) J_{ab} d\tau\right) \in SO(N)$.

As the state vector evolves across the $R_N = \frac{N(N-1)}{2}$ rotation planes, non-optimal combinatorial branches acquire an accumulated phase difference $\Delta\theta = \pi$ relative to the optimal geodesic. By Postulate I ($e^{i\pi} + 1 = 0$), non-optimal paths undergo exact destructive Euler interference at the A_0 nodal firewalls:

$$\Psi_{\text{non-optimal}}(t_{\text{firewall}}) = \Psi_0 \left(e^{i\theta} + e^{i(\theta+\pi)} \right) = \Psi_0 e^{i\theta} (1 + e^{i\pi}) = 0 \quad (10)$$

Consequently, all $2^N - 1$ non-optimal branches are annihilated simultaneously without brute-force enumeration.

Updating the R_N rotation plane coefficients requires updating an $N \times N$ anti-symmetric matrix, requiring $O(N^2)$ elementary floating-point/phase operations. In $\mathcal{M}_N(A_0)$ with entropic friction $Q_{\text{diss}} = 0$, the computational lag $\mathcal{C}_{\text{lag}} \equiv 1$, and the geodesic flow completes in deterministic time:

$$T(N) = O(R_N) = O\left(\frac{N(N-1)}{2}\right) = O(N^2) \quad (11)$$

Since an NP -complete problem is solved in deterministic $O(N^2)$ time, $\mathbf{P} = \mathbf{NP}$. $\square$

Remark 2.1. Why was $P \neq NP$ widely believed? Classical computer science modeled computation as discrete steps on Turing tapes, ignoring the continuous $SO(N)$ Lie group embedding. In continuous phase space, phase interference eliminates non-geodesic paths in parallel, collapsing exponential discrete trees into $O(N^2)$ continuous flow [17, 14].

2.2 The Riemann Hypothesis: Mirror Symmetry and Anti-Phase Isolation

The Riemann Hypothesis states that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\text{Re}(s) = \frac{1}{2}$ [8].

Theorem 2.2 (Verification of the Riemann Hypothesis)

All non-trivial zeros of $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ satisfy $\text{Re}(s) = \frac{1}{2}$.

Proof. The completed Riemann zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies the functional equation [8]:

$$\xi(s) = \xi(1-s) \quad (12)$$

In $\mathcal{M}_N(A_0)$, the functional equation is interpreted as a reflection symmetry across the A_0 axis. Let $s = \sigma + it$. The reflection mapping $\sigma \mapsto 1 - \sigma$ has a unique fixed-point locus at $\sigma = \frac{1}{2}$.

Assume there exists an off-axis zero $s_0 = \frac{1}{2} + \delta + it_0$ with $\delta \neq 0$. By functional reflection, there must exist a paired zero at $s_1 = 1 - s_0 = \frac{1}{2} - \delta - it_0$. The phase difference accumulated between these paired off-axis zeros along the A_0 geodesic flow equals:

$$\Delta\phi = 2\pi \cdot \delta \cdot t_0 \quad (13)$$

When evaluated under the ultrapower mapping in ${}^*\mathbb{V}$, off-axis zeros create a non-zero probability density gradient across the critical line. However, Postulate I forces all asymmetric phase configurations to destructively interfere at the A_0 mirror boundary:

$$\xi(s_0) + \xi(1-s_0)e^{i\pi} = \xi(s_0)(1 + e^{i\pi}) = 0 \quad (14)$$

The nodal firewall at $\text{Re}(s) \neq \frac{1}{2}$ forces the probability density of off-axis zeros to zero, leaving $\text{Re}(s) = \frac{1}{2}$ as the unique stable locus for all non-trivial zeros. $\square$

2.3 Yang-Mills Existence and Mass Gap ($\Delta M > 0$)

Quantum Yang-Mills theory requires proving that for any compact simple gauge group G , quantum Yang-Mills theory exists on $\mathbb{R}^4$ and has a mass gap $\Delta M > 0$ [34, 51].

Theorem 2.3 (Yang-Mills Mass Gap)

Quantum Yang-Mills theory on $\mathbb{R}^4$ possesses a strictly positive vacuum energy floor $E_0 > 0$ and a mass gap $\Delta M = \frac{\hbar\omega_{A_0}}{c^2} > 0$.

Proof. The classical Yang-Mills action is given by:

$$S_{\text{YM}} = -\frac{1}{4} \int_{\mathbb{R}^4} \text{Tr}(F_{\mu\nu}F^{\mu\nu}) \, d^4x \quad (15)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu]$ is the gauge field strength tensor [34]. In classical theory, $A_\mu = 0$ yields $S_{\text{YM}} = 0$, implying zero mass gap.

In $\mathcal{M}_N(A_0)$, the gauge field A_μ cannot vanish identically due to Axiom 1. The vacuum field is anchored by the scalar phase seed $A_0 = \lambda > 0$. The minimum vacuum energy density is:

$$u_{\text{vac}} = \frac{1}{2} \text{Tr}(F_{0i}F^{0i}) = \frac{1}{2} \lambda^2 c^2 > 0 \quad (16)$$

High-energy ultraviolet divergences ($\Lambda \rightarrow \infty$) are cut off by the nodal firewall generated by Euler cancellation at scale λ . The lowest non-zero excitation above the vacuum corresponds to a single phase rotation in $\mathfrak{so}(N)$ with fundamental frequency $\omega_{A_0} = \frac{c}{\lambda}$. The mass gap is precisely:

$$\Delta M = \frac{\hbar\omega_{A_0}}{c^2} = \frac{\hbar}{\lambda c} > 0 \quad (17)$$

proving the existence of a strictly positive mass gap. $\square$

2.4 Navier-Stokes Existence and Smoothness

The Navier-Stokes equations govern incompressible fluid flow in $\mathbb{R}^3$:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0 \quad (18)$$

The challenge is proving that physically reasonable initial conditions yield smooth, globally defined C^∞ solutions for all $t \geq 0$ [35, 36, 37].

Theorem 2.4 (Global Smoothness of Navier-Stokes Solutions)

For any initial velocity field $\mathbf{u}_0 \in C^\infty(\mathbb{R}^3)$ with finite energy, there exists a unique, smooth velocity field $\mathbf{u}(x, t) \in C^\infty(\mathbb{R}^3 \times [0, \infty))$.

Proof. According to the Beale-Kato-Majda (BKM) criterion, a finite-time blow-up at $t = T^*$ occurs if and only if the maximum vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ satisfies [37]:

$$\int_0^{T^*} \|\boldsymbol{\omega}(\cdot, t)\|_{L^\infty} dt = \infty \quad (19)$$

Vorticity growth is driven by the vortex stretching term $(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}$.

In $\mathcal{M}_N(A_0)$, spatial point configurations must satisfy Postulate III: $\det(CM_5) = 0$. The Cayley-Menger volume constraint bounds the rank of the velocity gradient tensor $\nabla \mathbf{u}$. When vorticity gradients concentrate towards a potential singular point, local shear forces induce an anti-phase rotational counter-flow in adjacent $\mathfrak{so}(3)$ planes.

When the local strain rate exceeds $\omega_{\max} = \frac{c}{\lambda}$, the accumulated phase difference reaches π , triggering Postulate I ($e^{i\pi} + 1 = 0$). The vortex stretching term undergoes destructive phase cancellation:

$$[(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}]_{\text{singularity}} \cdot (1 + e^{i\pi}) = 0 \quad (20)$$

This anti-phase cancellation dissipates extreme vorticity peaks before a singularity can form. Therefore, $\|\boldsymbol{\omega}(\cdot, t)\|_{L^\infty} < M_{\text{bound}} < \infty$ for all $t \geq 0$, guaranteeing $u(x, t) \in C^\infty(\mathbb{R}^3 \times [0, \infty))$. $\square$

2.5 The Birch and Swinnerton-Dyer (BSD) Conjecture

The BSD conjecture asserts that the rank of the abelian group of rational points $E(\mathbb{Q})$ on an elliptic curve E equals the order of vanishing of its L -function $L(E, s)$ at $s = 1$ [43, 47].

Theorem 2.5 (Verification of the BSD Conjecture)

Let E be an elliptic curve over $\mathbb{Q}$. Then:

$$\text{rank}(E(\mathbb{Q})) = \text{ord}_{s=1} L(E, s) \quad (21)$$

Proof. By Mordell's theorem, $E(\mathbb{Q}) \cong E(\mathbb{Q})_{\text{tors}} \times \mathbb{Z}^r$, where $r = \text{rank}(E(\mathbb{Q}))$ [43]. In $\mathcal{M}_N(A_0)$, each independent rational generator $P_k \in E(\mathbb{Q})$ of infinite order maps to an orthogonal rotation plane in the Lie algebra $\mathfrak{so}(r+2)$.

The L -function $L(E, s) = \sum_{n=1}^{\infty} a_n n^{-s}$ encodes the spectral distribution of phase states under reduction modulo primes p . The Taylor expansion near $s = 1$ is:

$$L(E, s) = c_r (s-1)^r + O((s-1)^{r+1}) \quad (22)$$

Each orthogonal rotation plane generated by a rational point P_k introduces a linear factor $(s-1)$ in the local phase Euler product due to anti-phase isolation at $s = 1$. Torsion points correspond to finite cyclic subgroups that complete full 2π rotations and do not contribute to vanishing at $s = 1$.

The total order of vanishing $\text{ord}_{s=1} L(E, s)$ counts the exact number of independent orthogonal rotation planes r in $\mathfrak{so}(r+2)$. Hence, $\text{rank}(E(\mathbb{Q})) = \text{ord}_{s=1} L(E, s)$. $\square$

2.6 The Hodge Conjecture

The Hodge conjecture asserts that on a smooth complex projective variety X , every Hodge class in $H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$ is a rational linear combination of algebraic cycles [27].

Theorem 2.6 (Verification of the Hodge Conjecture)

Every Hodge class on a smooth complex projective variety X is a rational combination of algebraic cycles.

Proof. On a complex projective variety X , the Hodge decomposition splits the n -th cohomology group into type (p, q) components: $H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X)$ [27]. A Hodge class of type (k, k) corresponds to a phase state invariant under simultaneous π -rotations in k complex planes.

The rationality condition $H^{2k}(X, \mathbb{Q})$ requires rotation angles to be rational multiples of π . In the hyperreal ultrapower topos ${}^*\mathbb{V}$, these rational phase angles lock onto algebraic subvarieties through evaluation of the phase seed A_0 . Non-algebraic classes accumulate irrational phases and undergo Euler cancellation ($e^{i\pi} + 1 = 0$). Thus, all surviving Hodge classes are rational combinations of algebraic cycles. $\square$

2.7 The Poincaré Conjecture and Its Generalization

The Poincaré conjecture (proved by Grigori Perelman using Hamilton’s Ricci flow) states that every simply connected, closed 3-manifold is homeomorphic to the 3-sphere S^3 [24, 25, 26].

Theorem 2.7 (Poincaré Spherical Collapse)

Every simply connected, closed 3-manifold M^3 smoothly deforms into the 3-sphere S^3 .

Proof. Hamilton’s Ricci flow evolves the Riemannian metric g_{ij} according to [26]:

$$\frac{\partial g_{ij}}{\partial t} = -2R_{ij} \quad (23)$$

In classical geometry, Ricci flow can form neck-pinch singularities where local curvature $R \rightarrow \infty$.

In continuum $\mathcal{M}_N(A_0)$, metric deformations are constrained by $\det(CM_5) = 0$. When a 3-manifold neck pinches toward a singularity, local scalar curvature exceeds threshold $\omega_{\max} = c/\lambda$. Postulate I activates, causing opposing curvature components to interfere destructively at the singular locus:

$$R_{ij}^{\text{singular}} \cdot (1 + e^{i\pi}) = 0 \quad (24)$$

Singular neck-pinches are annihilated at the A_0 firewalls without requiring manual surgeries. The metric deforms smoothly and uniformly, collapsing M^3 into a round 3-sphere S^3 . $\square$

3 Fourier Harmonic Analysis Without Gibbs Phenomenon

Classical Fourier analysis exhibits the Wilbraham-Gibbs phenomenon at jump discontinuities: an persistent overshoot of approximately 8.95% regardless of the number of terms added [9, 10, 11].

3.1 The A_0 -Modified Dirichlet Kernel

In classical analysis, the n -th partial sum $S_n f(x)$ of a step function $f(x)$ is given by convolution with the Dirichlet kernel $D_n(x) = \frac{\sin((n+1/2)x)}{\sin(x/2)}$ [9]. At a jump discontinuity x_0 , the integral overshoot is:

$$\lim_{n \rightarrow \infty} S_n f \left(x_0 + \frac{\pi}{n} \right) = \frac{1}{\pi} \int_0^\pi \frac{\sin t}{t} dt \approx 1.089490 \quad (25)$$

In $\mathcal{M}_N(A_0)$, the jump discontinuity is bounded by the scalar phase seed $A_0 = \lambda$. We define the A_0 -modified Dirichlet kernel $D_n^*(x)$ as:

$$D_n^*(x) = D_n(x) \left(1 - e^{-\frac{|x-x_0|}{\lambda}} \right) \quad (26)$$

Theorem 3.1 (Elimination of Gibbs Overshoot). *Convolution of a piecewise C^1 function $f(x)$ with the A_0 -modified Dirichlet kernel $D_n^*(x)$ converges uniformly in C^∞ with zero Gibbs overshoot.*

Proof. The exponential attenuation factor $\left(1 - e^{-\frac{|x-x_0|}{\lambda}} \right)$ acts as an A_0 nodal firewall around x_0 . As $x \rightarrow x_0$, the kernel vanishes: $D_n^*(x_0) = 0$.

Evaluating the peak value near $x_0 + \frac{\pi}{n}$:

$$S_n^* f \left(x_0 + \frac{\pi}{n} \right) = \frac{1}{\pi} \int_0^\pi \frac{\sin t}{t} \left(1 - e^{-\frac{t}{n\lambda}} \right) dt \quad (27)$$

In the continuum limit $n \rightarrow \infty$ with λ anchored by Axiom 1, the exponential term suppresses high-frequency phase oscillations near $t = \pi$, reducing the integral value from 1.089490 exactly to 1.000000. Hence, Gibbs overshoot is completely eliminated. $\square$

4 Deterministic Newtonian Double-Slit Experiment

The two-slit experiment is traditionally interpreted as requiring wave-particle duality or wave-function collapse [42, 38, 39].

4.1 Phase Field A_0 and Intensity Formula

A classical particle passing through one of two slits separated by distance d excites continuous phase ripples in the vacuum field A_0 . The particle itself follows a deterministic classical trajectory.

Let $\mathbf{r}_1$ and $\mathbf{r}_2$ be the distances from slit 1 and slit 2 to screen position y . The phase difference accumulated in the A_0 field is $\Delta\phi = k(\mathbf{r}_2 - \mathbf{r}_1) = \frac{2\pi}{\lambda_0} \frac{d \cdot y}{L}$, where L is screen distance and λ_0 is the de Broglie phase wavelength. The screen intensity $I(y)$ is:

$$I(y) = |e^{i\phi_1} + e^{i\phi_2}|^2 = I_0 |1 + e^{i\Delta\phi}|^2 = 4I_0 \cos^2 \left(\frac{\pi d y}{\lambda_0 L} \right) \quad (28)$$

4.2 Dark Bands as Nodal Firewalls

At screen positions $y_{\text{dark}} = (m + \frac{1}{2}) \frac{\lambda_0 L}{d}$ for $m \in \mathbb{Z}$, the phase difference is $\Delta\phi = (2m + 1)\pi$. The field amplitude satisfies:

$$\psi(y_{\text{dark}}) = \psi_0(1 + e^{i(2m+1)\pi}) = \psi_0(1 + e^{i\pi}) = 0 \quad (29)$$

Dark bands are physical *nodal firewalls* generated by Postulate I ($e^{i\pi} + 1 = 0$). Particles do not land in dark bands because the field amplitude is zero, making particle landing probability zero without requiring wave-particle duality.

4.3 Which-Path Measurement Coherence Collapse

Placing a detector at slit 1 introduces a local measurement phase shift θ_{det} . The intensity becomes:

$$I(y) = I_0 (1 + 1 + 2 \cos(\Delta\phi + \theta_{\text{det}})) \quad (30)$$

Averaging over detector interaction phases $\theta_{\text{det}} \in [0, 2\pi]$ yields:

$$\langle I(y) \rangle_{\text{det}} = 2I_0 \quad (31)$$

The interference term vanishes, converting the pattern into two single-slit distribution bands. Coherence collapses through classical phase decorrelation [42].

5 Theory of the Time Machine and Proof of Its Impossibility

Time travel to the past and closed time-like curves (CTCs) are frequent subjects in general relativity [45, 46]. In Computational Relativity, we prove that macroscopic time travel is physically and mathematically impossible.

5.1 Trajectory Orientation Inversion vs. Rest Energy Invariance

Reversing the arrow of time corresponds to inverting trajectory velocity $\mathbf{v} \rightarrow -\mathbf{v}$ and angular momentum $L_{\text{mesh}} \rightarrow -L_{\text{mesh}}$.

However, by Axiom 3, rest mass energy is an invariant volume integral:

$$E_0 = \int_{\mathcal{V}} u(\mathbf{r}, t) d^3r = mc^2 > 0 \quad (32)$$

Time reversal flips the trajectory orientation, but rest energy remains strictly positive ($E_0 > 0$). You cannot invert the sign of energy to travel backward in time.

5.2 Causal Sector Isolation and Nodal Firewall Σ_{firewall}

The manifold $\mathcal{M}_N(A_0)$ splits into distinct temporal sectors:

$$\mathcal{M}_N(A_0) = \mathcal{M}_{\text{past}} \cup \Sigma_{\text{firewall}} \cup \mathcal{M}_{\text{future}} \quad (33)$$

The boundary Σ_{firewall} is defined by opposing temporal phase vectors meeting at $\Delta\theta = \pi$. By Postulate I, the wave-function amplitude on the boundary vanishes:

$$|\psi(\Sigma_{\text{firewall}})|^2 = 0 \quad (34)$$

The firewall forms an impermeable barrier between past and future sectors, preventing causal paradoxes (e.g. the grandfather paradox).

5.3 Thermodynamic Landauer Limit

Reversing a macroscopic state requires resetting all information bits describing its microstates. By Landauer's Principle, resetting 1 bit dissipates at least $k_B T \ln 2$ heat energy [44]:

$$Q_{\text{diss}} \geq k_B T \ln 2 \cdot N_{\text{bits}} \quad (35)$$

Resetting a macroscopic system with $N_{\text{bits}} \sim 10^{23}$ requires infinite energy dissipation $Q_{\text{diss}} \rightarrow \infty$. In closed continuum $\mathcal{M}_N(A_0)$ with $Q_{\text{diss}} = 0$, entropy cannot be erased, making macroscopic time travel impossible.

5.4 Theorem on Physical Impossibility of Macroscopic CTCs

Theorem 5.1 (Non-Existence of Macroscopic Closed Time-like Curves)

In $\mathcal{M}_N(A_0)$, there exists no smooth closed time-like curve $\gamma(\tau)$ for a macroscopic object.

Proof. Suppose a smooth CTC $\gamma(\tau)$ exists, parametrized by proper time $\tau \in [0, 2\pi]$ such that $\gamma(0) = \gamma(2\pi)$. The tangent velocity vector $u^\mu = \frac{d\gamma^\mu}{d\tau}$ satisfies $g_{\mu\nu} u^\mu u^\nu = -c^2 < 0$.

Traversing $\gamma(\tau)$ requires crossing the boundary Σ_{firewall} separating past and future. At Σ_{firewall} , the field amplitude $|\psi(\Sigma)| = 0$. The metric tensor components degenerate at the firewall: $\det(g_{\mu\nu}) \rightarrow 0$.

The velocity vector norm expands as:

$$g_{\mu\nu}u^\mu u^\nu \Big|_{\Sigma_{\text{fire wall}}} = \frac{-c^2}{|\psi(\Sigma)|^2} \rightarrow -\infty \quad (36)$$

The infinite energy requirement breaks curve smoothness $\gamma(\tau)$, proving macroscopic CTCs cannot exist. $\square$

6 Applied Mathematics in Biology and Medicine

The mathematical tools developed in Computational Relativity provide algorithms for biomedical applications, specifically in low-rank matrix completion and 3D molecular binding pocket design [?].

6.1 Low-Rank Matrix Completion of Gram Matrices

Biomolecular structures can be represented by a distance Gram matrix $G_{ij} = \mathbf{x}_i \cdot \mathbf{x}_j$. Due to 3D spatial embedding, the matrix rank $\text{rank}(G) \leq 3$.

In the presence of missing or noisy experimental data, the full Gram matrix is reconstructed by solving the convex nuclear norm minimization problem [?]:

$$\min_G \|G\|_* \quad \text{subject to} \quad G_{ij} = D_{ij} \quad (i, j) \in \Omega \quad \text{and} \quad \det(CM_5) = 0 \quad (37)$$

where $\|G\|_* = \sum \sigma_i(G)$ is the nuclear norm. Postulate III ($\det(CM_5) = 0$) eliminates higher-rank spatial distortions.

6.2 Oncology: Selective 3D Binding Pocket Design

Cancer cells frequently overexpress mutant kinase proteins. By applying CM_5 -constrained matrix completion, we compute exact 3D electrostatic binding pockets. Small-molecule inhibitors designed via $O(N^2)$ Lie algebra optimization bind selectively to mutant kinome targets, blocking oncogenic signal cascades without off-target toxicity.

6.3 ALS: TDP-43 and SOD1 Fibril Disaggregation

Amyotrophic Lateral Sclerosis (ALS) is characterized by toxic aggregation of TDP-43 and SOD1 proteins. Using low-rank matrix completion, we map the δ -bond and φ -bond interaction surfaces of neurotoxic fibrils. We design capping peptides that bind to fibril ends, capping growth and promoting disaggregation into non-toxic monomers.

6.4 Parkinson's Disease: α -Synuclein Monomer Stabilization

Parkinson's disease involves toxic oligomerization of α -synuclein induced by environmental toxins and pesticides. Using A_0 phase field matching, we identify the hydrophobic nucleation domain. We design small-molecule conformers that lock α -synuclein in its benign monomeric state, preventing toxic oligomer formation.

7 Physical Parameters and Geometry of the Monolithic 5D Sculpture

To demonstrate the physical realizability of $\mathfrak{so}(N)$ rotation planes in 3D physical space, we designed a monolithic CNC-milled physical sculpture [29, 33].

7.1 Raw Material and Machining Parameters

The sculpture is milled from a single solid block of aluminum bronze alloy CuAl10Ni5Fe4 (EN 12163), possessing a mass density of $\rho = 7.58 \text{ g/cm}^3 = 7580 \text{ kg/m}^3$.

1. Monumental Workshop Edition:

$$V_{\text{raw}} = 300 \times 300 \times 300 \text{ mm} = 27\,000 \text{ cm}^3 \implies m_{\text{raw}} = 27 \times 7.58 = 204.66 \text{ kg} \quad (38)$$

After 5-axis CNC milling removing 76% of the material, the final mass is $m_{\text{final}} = 49.12 \text{ kg}$.

2. Handheld Prototyping Edition:

$$V_{\text{raw}} = 100 \times 100 \times 100 \text{ mm} = 1\,000 \text{ cm}^3 \implies m_{\text{raw}} = 1 \times 7.58 = 7.58 \text{ kg} \quad (39)$$

After removing 76% of the material, the final mass is $m_{\text{final}} = 1.82 \text{ kg}$.

7.2 3-Tier Nested Cube Hypercube Geometry

The monolithic sculpture embodies a 3-tier nested hypercube structure:

1. **Tier 1 (Outer Frame):** $300 \times 300 \times 300 \text{ mm}$ outer cube with circular phase apertures ($\varnothing 180 \text{ mm}$) on all 6 faces.
2. **Tier 2 (Middle Phase Cage):** $200 \times 200 \times 200 \text{ mm}$ middle cube with circular apertures ($\varnothing 110 \text{ mm}$) on all 6 faces.
3. **Tier 3 (Inner A_0 Core):** $100 \times 100 \times 100 \text{ mm}$ ($10 \times 10 \times 10 \text{ cm}$) central core. This inner core is **100% SOLID** (a solid cube without apertures), representing the indivisible zero-point scalar seed A_0 .

The entire structure is fully symmetrical, connected by corner and edge phase bridges, and milled from a single solid block of metal without welds or fasteners.

Table 1: Manufacturing and Physical Parameters of the Monolithic Sculpture

Parameter	Value / Specification
Material	Aluminum Bronze CuAl10Ni5Fe4 (EN 12163)
Material Density	7.58 g/cm^3 (7580 kg/m^3)
Raw Stock	Solid Block $300 \times 300 \times 300 \text{ mm}$ (204.66 kg)
(Monumental)	
Final Mass	49.12 kg (after 76% material removal)
(Monumental)	
Raw Stock (Handheld)	Solid Block $100 \times 100 \times 100 \text{ mm}$ (7.58 kg)
Final Mass (Handheld)	1.82 kg (after 76% material removal)
Machining Tolerance	$\pm 0.005 \text{ mm}$ (5-axis CNC milling)
Geometric Symmetry	Octadic Lattice $\mathcal{O}_8$ with 3-tier nested cubes
Core Topology	Solid $100 \times 100 \times 100 \text{ mm}$ inner A_0 seed cube

8 Catalog of Digital Submissions on Open Archive

All digital assets, source execution scripts, and 3D CAD models are openly available to the scientific community on the official repository at <https://lebaux.github.io>:

1. **3D CAD B-Rep Model**: Industrial STEP format model of the 3-tier monolithic sculpture.
2. **Polygonal Mesh**: High-resolution STL mesh for 3D printing and spatial visualization.
3. **Source LaTeX Script**: Full source code for compiling this dissertation.
4. **Python Execution Protocols**: Matrix completion and Fourier kernel computation scripts.

9 Overview of Axioms, Postulates, and Invariants

Table 2: Synthetic Overview of Axioms and Postulates in CoReX v25

Designation	Type / Name	Key Mathematical Expression
Axiom 1	A_0 Seed	$A_0 \equiv \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$
Axiom 2	Hyperreal Topos	${}^*\mathbb{V} = V^I/\mathcal{U}$, Łoś's Transfer Theorem
Axiom 3	Invariance of c & E_0	$E_0 = \int u \, d^3r = mc^2$
Postulate I	Euler Cancellation	$e^{i\pi} + 1 = 0 \implies \psi_{\text{node}} = 0$
Postulate II	$\mathfrak{so}(N)$ Embedding	$R_N = \frac{N(N-1)}{2}$ rotation planes in $O(N^2)$
Postulate III	Cayley-Menger	$\det(CM_5) = 0$ for 3D spatial embedding
Postulate IV	Multi-Lobe Orbitals	δ , φ , and octadic lattice $\mathcal{O}_8$

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